Quantum Bitcoin : A Post-Quantum Peer-to-Peer Electronic Cash System

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Abstract

This paper presents a peer-to-peer electronic cash architecture designed to mitigate cryptographic vulnerabilities introduced by scalable quantum computing. Extant networks relying on the Elliptic Curve Digital Signature Algorithm (ECDSA) face systemic exposure as physical qubit capacities expand. Shor's algorithm solves the elliptic curve discrete logarithm problem within polynomial time, converting the statistical ten-minute mempool propagation interval into an exploit window for quantum-enabled transaction front-running.

To resolve this vulnerability, the Quantum Bitcoin framework eliminates ECDSA at the consensus layer, natively implementing the Module Lattice-Based Digital Signature Standard (ML-DSA). To counter the data inflation and verification overhead inherent to lattice-based primitives, we introduce serialized state isolation, post-quantum witness decoupling (PQ-SegWit), and a 128-bit fixed-point continuous difficulty adjustment mechanism. The network initializes via an independent genesis block with a strict mathematical supply ceiling of 21,000,000 units, preserving decentralized asset ownership and validation in a post-quantum environment.

Chapter 1: Cryptographic Boundaries and Time Exposure of Extant Networks

1.1 Algebraic Vulnerabilities of Elliptic Curves

The ownership verification models of conventional peer-to-peer electronic cash systems depend on the secp256k1 elliptic curve standard. The security assumptions of this framework rely on the non-polynomial time complexity required for classical architectures to solve the Elliptic Curve Discrete Logarithm Problem (ECDLP). Shor's algorithm, by exploiting quantum state superposition, reduces this complexity class to polynomial time. Under the execution of a Cryptographically Relevant Quantum Computer (CRQC), the relationship between an ECDSA public key and its corresponding private key becomes analytically reversible.

1.2 The Mempool Propagation Latency Window

Standard protocols frequently operate under the assumption that address reuse avoidance constitutes a sufficient defense against public-key exploitation. This assumption fails to account for the physical propagation delays inherent to peer-to-peer routing networks. Upon transaction initialization, the plaintext ECDSA signature and the unhashed public key are exposed to the global unconfirmed transaction pool (Mempool). Prior to the inclusion of the transaction into a block via the proof-of-work mechanism, an exposure window exists with a statistical expectation of ten minutes.

Within this temporal window, an adversary equipped with quantum computing capacity can extract the public key from the broadcast, derive the private key, and construct a competing transaction featuring an elevated transaction fee (Replace-by-Fee). Under rational economic incentives, network validation nodes prioritize the transaction offering higher fee density. Consequently, the attacker can redirect the underlying assets without maintaining a 51% consensus hash rate advantage. Long-term network stability therefore necessitates the deployment of signature schemes devoid of algebraic geometric periodicity directly within the consensus path, eliminating the analytical vector for private key derivation during transmission.

Chapter 2: Base-Layer Consensus Reconstruction and Cryptographic Migration

2.1 Post-Quantum Public Key Infrastructure

To establish cryptographic resilience against quantum computation, the network removes ECDSA from the base consensus Layer 1, integrating the Module Lattice-Based Digital Signature Standard (specifically ML-DSA-65). The cryptographic hardness of lattice-based primitives is rooted in the shortest vector problem (SVP) across high-dimensional vector spaces, an algebraic structure that lacks the geometric periodicity exploitable by Shor's algorithm. By embedding ML-DSA-65 directly into the transaction validation pipeline, the protocol mathematically invalidates private key derivation from broadcast public data.

2.2 Deterministic Difficulty Adjustment Model

Discrete difficulty adjustment mechanisms introduce block-time variances and temporal manipulation vulnerabilities under non-linear hash rate fluctuations. This framework deploys the ASERTi3-2d continuous difficulty adjustment algorithm. To eliminate non-deterministic rounding errors across heterogeneous hardware architectures during floating-point operations, the underlying execution engine utilizes a 128-bit fixed-point integer arithmetic model. The network computes mathematical feedback adjustments for every sequence initialization, ensuring the empirical block interval systematically converges toward a predefined constant parameter (600 seconds) within highly dynamic computational environments.

2.3 Independent Genesis and Supply Constants

The state space initializes via a discrete Genesis Block, containing no historical ledger inheritance or external state mappings. Asset generation occurs exclusively through verified proof-of-work hash collisions satisfying the active difficulty target. The consensus parameters dictate a terminal supply limit of 21,000,000 units alongside a monotonically decaying emission curve indexed to block height. This mechanism establishes objective distribution parameters and guarantees a trustless mathematical scarcity over long horizons.

Chapter 3: State Determinism and Witness Decoupling

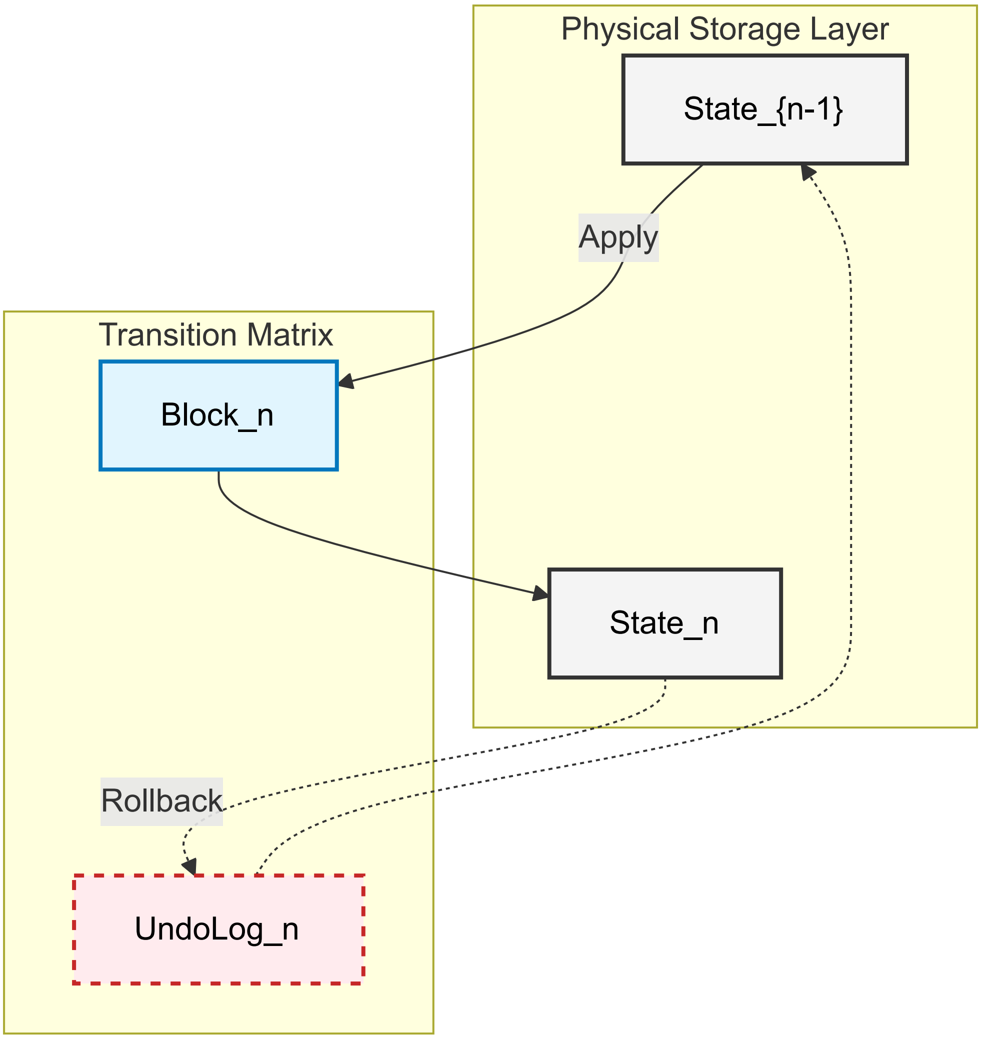
3.1 State Determinism and Temporal Reversibility

Peer-to-peer networks exhibit concurrent block propagation and deep chain reorganization. Conventional node architectures executing state overwrites face data inconsistency risks during abrupt external interruptions such as hardware power failures.

This protocol models the Unspent Transaction Output (UTXO) set as a strict deterministic finite state machine. For any state transition driven by a new block B_n, the validation engine requires the construction of a corresponding inverse transformation set, designated as the Undo Log (U_n), prior to execution. When a longest-chain reorganization event occurs, validation nodes execute rollback operations by applying the inverse function along the chronological axis:

$$State_{n-1} = State_n \ominus U_n$$

This mechanism yields temporal reversibility with an $O(1)$ computational complexity profile. Through systematic serial isolation, the ledger maintains mathematical consistency independent of topological network reconfigurations.



(Figure 1: State Machine Bi-directional Transition and Undo Log Isolation)

3.2 Post-Quantum Witness Separation (PQ-SegWit)
Module lattice signature schemes inherently increase the data payload required per transaction input. Unconstrained ledger growth elevates the physical storage thresholds required to operate a validation node, threatening decentralization.

The core state transitions (Inputs and Outputs) and their associated cryptographic proofs (Witness data) possess asymmetric utility profiles over time. The architecture enforces a physical bifurcation of these data structures. Once a block achieves sufficient confirmation depth within the main chain—indicating that its state alterations are permanently bound by cumulative proof-of-work—validation nodes prune the historical witness data from local operational storage:

$$\text{If } Depth(Block) > K, \ \text{Prune}(Witness_{data}) \rightarrow \text{True}$$

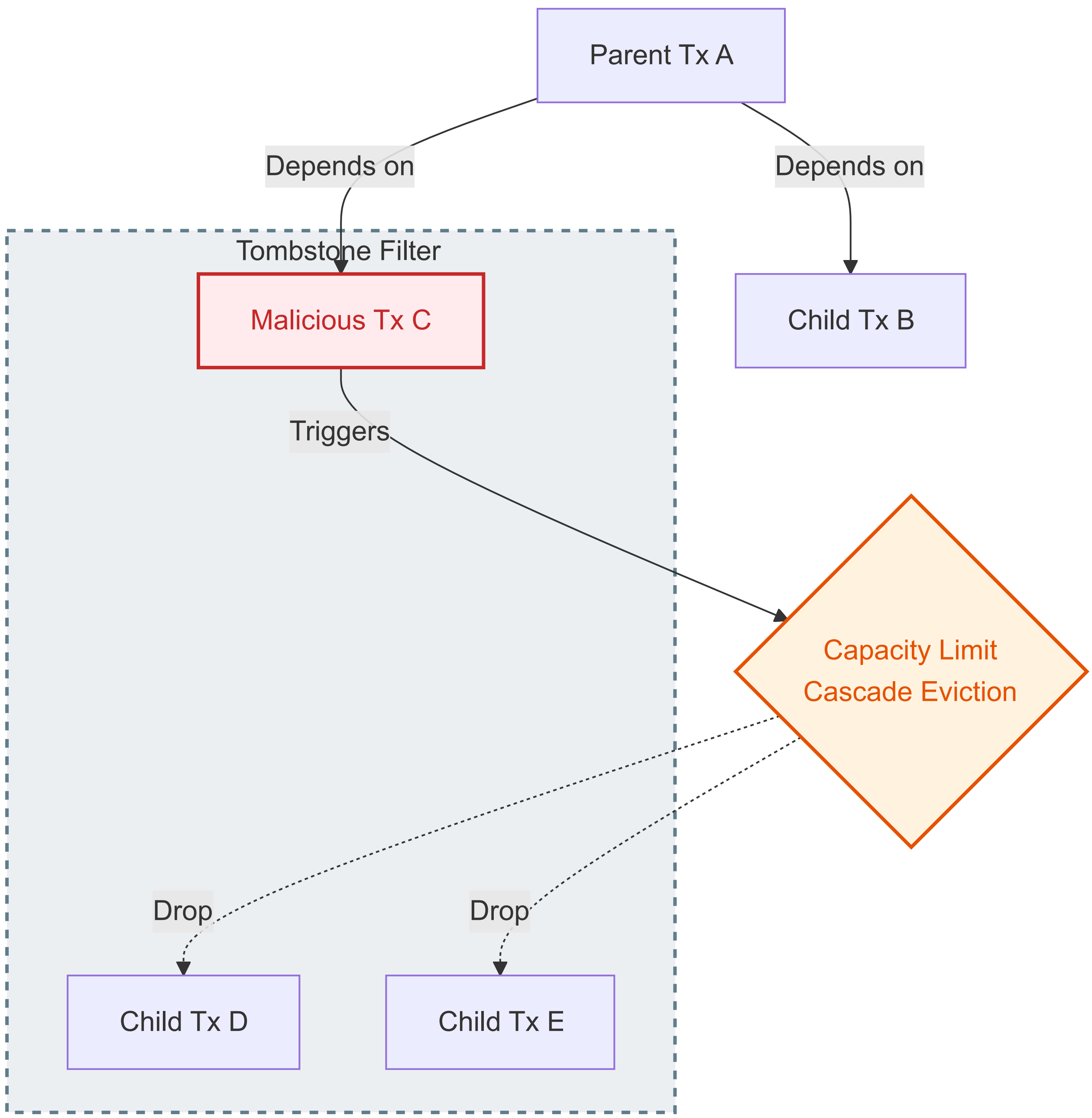
This decoupling guarantees that the cryptographic state mutations remain permanently recorded while the historical proof payloads are vacated post-consensus. This optimization bounds the spatial expansion rate of the post-quantum network near the theoretical limits of classical systems.

Chapter 4: Mempool Topology and Compact Network Propagation

4.1 Directed Acyclic Graph (DAG) and Resource Exhaustion Defense

Unconfirmed transactions propagating through a peer-to-peer network expose individual nodes to memory exhaustion vectors via high-frequency micro-transaction flooding (Dust Flooding).

To mitigate this attack vector, the mempool architecture models unconfirmed states as a strict Directed Acyclic Graph (DAG), where transaction dependencies define directed edges. When node memory utilization crosses a predefined safety threshold, the eviction engine initiates a cascade deletion based on economic density (Fee-to-Weight ratio). The removal of a parent node automatically invalidates and evicts all dependent child transactions across the DAG topology. Concurrently, evicted transaction hashes are committed to a localized cryptographic filter (Tombstone set), systematically blocking the re-propagation of invalid payloads at the network boundary.



(Figure 2: DAG Mempool Dependency Tree and Cascade Eviction Architecture)

4.2 Compact Routing and Truth Isolation Barriers
The physical dimensions of post-quantum cryptographic keys introduce propagation latency overheads if blocks are transmitted in full. The protocol deploys Compact Block Routing natively. Nodes restrict initial block broadcasts to structural skeletons containing SipHash short identifiers (Short IDs). Receiving nodes reconstruct the complete block payload utilizing matching transactions already resident in their local mempools, minimizing cross-network bandwidth consumption.

Furthermore, during Initial Block Download (IBD), the node architecture enforces an isolated view model. Incoming historical blocks that have not cleared full cryptographic validation are restricted to ephemeral memory spaces where their topological relations are mapped. Unverified state changes are physically barred from writing to the persistent local state database, isolating the node's true ledger state from potential contamination by adversarial or invalid chain forks.

Chapter 5: State Orthogonality and Asymmetric Time-Locks

5.1 Concurrent Verification Matrices Based on State Orthogonality
High-dimensional lattice cryptosystems impose measurable computational demands on validation hardware during single-signature verification operations. However, within the underlying UTXO model, discrete transaction sets that do not share identical inputs exhibit absolute mathematical orthogonality.

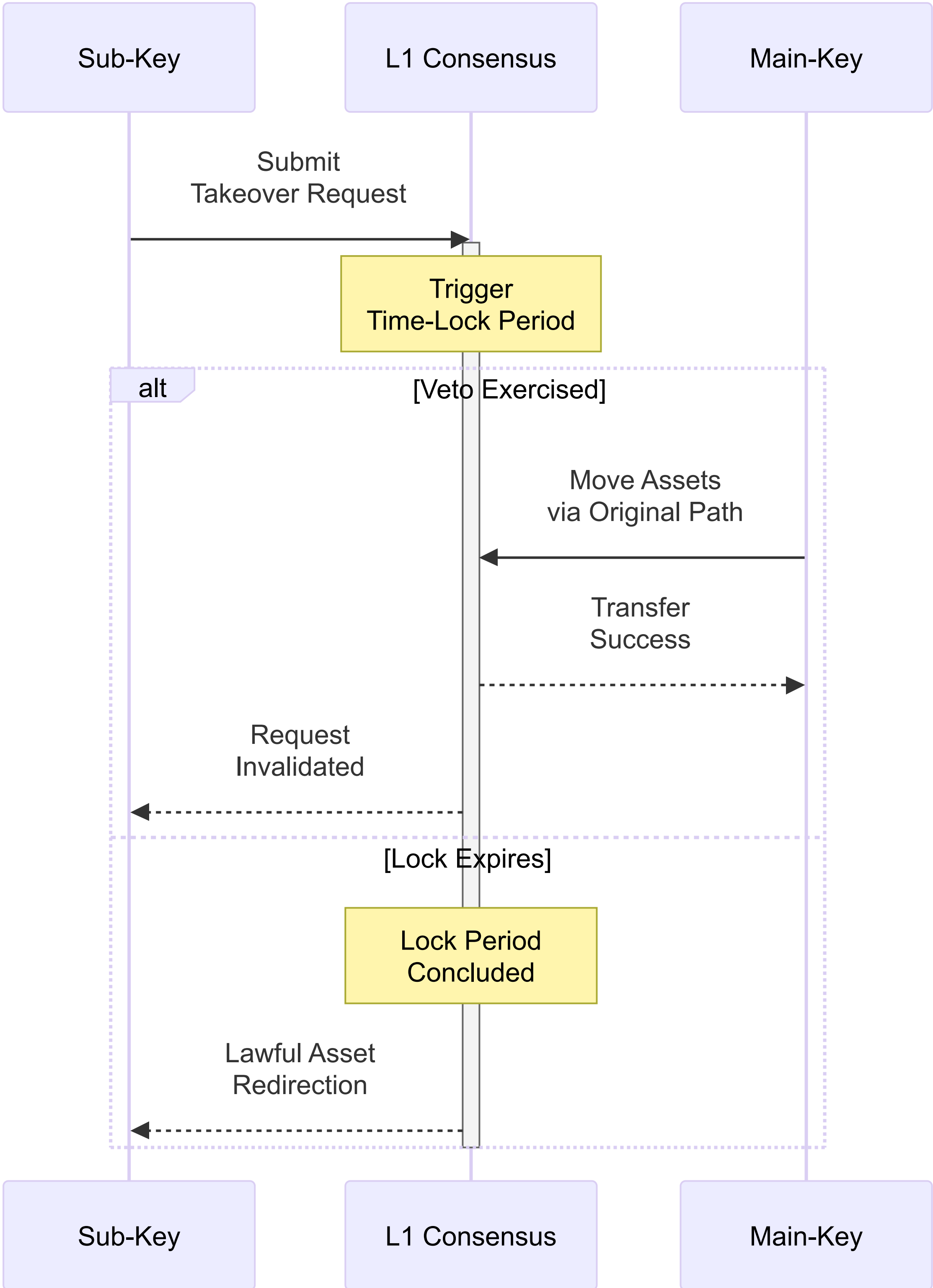
The validation engine leverages this property to bypass the constraints of sequential execution. Prior to block assembly and consensus propagation, nodes utilize an asynchronous parallel verification matrix to validate non-conflicting transaction branches concurrently across independent processing cores, caching the resultant boolean states. Upon block finalization and physical disk write, the engine references the pre-validated cache states. This design isolates dense cryptographic computation from the critical path of network synchronization, bounding block execution latency close to the physical input/output thresholds of the underlying hardware.

5.2 Time-Asymmetric Asset Recovery Mechanism
In post-quantum cryptographic frameworks, the irreversibility of lattice algebraic structures means the loss of a private key constitutes a catastrophic, unrecoverable event. To address this risk without compromising self-sovereignty, the protocol introduces an asymmetric time-locked asset recovery game model natively at Layer 1.

The underlying transaction output (TxOut) script structure permits the inclusion of a secondary public key hash paired with an incompressible absolute time-lock parameter (T_{lock}). Should the secondary key broadcast an operational takeover claim to the network, consensus rules enforce a cooling period of length T_{lock}. Within this temporal window:

$$T_{transfer} = T_{challenge} + T_{lock}$$

The primary key holder retains absolute priority to move the assets via the standard derivation path, which functions as an absolute veto. This mechanism operates independently of turing-complete smart contract runtimes, relying strictly on monotonically increasing block height to achieve equilibrium between catastrophic key recovery and absolute ownership security.



(Figure 3: Layer-1 Asymmetric Time-Lock Recovery Game Sequence)

Chapter 6: Monolithic Architecture and Stateless Verification Evolution

6.1 Monolithic Design and Global State Security

In addressing the throughput (TPS) challenges ubiquitous in decentralized networks, this system rejects scaling solutions such as tightly coupled cross-chain state migrations or sharding, which compromise global consensus security. The architecture adheres strictly to a monolithic design philosophy, ensuring that all state transitions are globally verified via cryptography by the L1 base layer, fundamentally eliminating the risk of atomicity failures introduced by cross-shard communication.

6.2 Layer-2 Scaling and Stateless Verification via ZK-STARKs

To counter long-term state bloat resulting from continuous operation while fiercely defending the Layer-1 monolithic baseline, the protocol outlines an evolutionary trajectory for a **Layer-2 scaling network** utilizing Zero-Knowledge Scalable Transparent Arguments of Knowledge (ZK-STARKs). ZK-STARKs operate without a trusted setup and are natively quantum-resistant, perfectly aligning with the system's security assumptions. By transforming the verification logic of historical states into succinct cryptographic STARK proofs off-chain, the L2 network will eventually permit edge nodes to verify the legitimacy of the global state without downloading the complete historical ledger, relying solely on mathematical proofs. This evolutionary route lays the theoretical and engineering foundation for the ultimate realization of the Stateless Verification paradigm, **ensuring the Layer-1 basechain remains unmodified, pure, and secure.**

Conclusion

This paper has demonstrated a base-layer architecture for a post-quantum peer-to-peer electronic cash system. By removing legacy elliptic curve constructs and natively integrating ML-DSA digital signatures, the framework provides verifiable resilience against Shor's algorithm. Utilizing directed acyclic graphs for mempool topology, serialized state isolation, and compact block routing, the network mitigates transaction propagation delays and memory exhaustion vectors. Addressing future scalability, the architecture adheres to a **strict Layer-1 monolithic design** and anchors an evolutionary **Layer-2 path** toward Stateless Verification via ZK-STARKs. Initialized by a discrete genesis block with a mathematically fixed 21,000,000 supply cap and continuous fixed-point difficulty targeting, the system preserves the decentralization of proof-of-work consensus while establishing long-term cryptographic stability.

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